

Debojyoti Ghosh

Computational Scientist | Large-scale physics simulation, numerical methods, and high-performance computing

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SUMMARY

Computational scientist with 13 years at DOE national laboratories building the large-scale simulation codes that physics research runs on. Over the past three years I wrote the particle-based cloud microphysics in a GPU-portable weather model, the implicit field-solver and preconditioning layer in an exascale particle-in-cell code, and the multi-disease machinery in an exascale agent-based epidemiological model — all in C++, CUDA and MPI, all merged upstream and in production use. My solvers also generate the training data behind several machine-learning and reduced-order-modeling studies, which I co-author. 29 journal articles, 1000+ citations, h-index 16.

SELECTED CONTRIBUTIONS

Super-Droplet cloud microphysics in ERF

2024 – present

Lead developer

- Wrote the particle-based Super-Droplet Method in [ERF](#), a performance-portable DOE weather code: the initial scheme, cold-cloud (ice) processes, adaptive-mesh support, terrain-following particle advection, and a structure-of-arrays particle container.
- Carried it to production on GPUs — CUDA, HIP and SYCL builds, single- and double-precision particle paths, and the CI matrix that keeps them working.

Implicit particle-in-cell solvers in WarpX

2024 – present

Solver and preconditioner developer

- Built the preconditioning and linear-solver layer for energy- and charge-conserving implicit electromagnetic PIC in [WarpX](#): a curl-curl operator preconditioner, sparse-matrix Jacobian assembly, and a PETSc SNES/KSP interface for preconditioned Newton-Krylov solves.
- This is what makes fully implicit dense-plasma simulation tractable, where explicit methods are limited by the plasma period and Debye length.

Multi-disease agent-based epidemiology in ExaEpi

2024 – present

Key developer

- Implemented simultaneous tracking of multiple diseases with co-immunity and co-susceptibility, capacity-limited hospitalization, and a medical workforce model in [ExaEpi](#), an exascale-capable agent-based model; refactored the core agent interaction kernels.

Simulation codes behind machine-learning and reduced-order modeling

2022 – present

- Supplied the physics solvers and generated the training data for reduced-order-modeling and ML studies — **HyPar** for electrostatic-plasma ROM, **ERF-SDM** for data-driven warm-rain microphysics — and co-authored the resulting papers (both 2026, below).
- *Early-stage exploration*: training conditioned U-Net and diffusion models in PyTorch to denoise particle-in-cell moments, with physics-informed loss terms and DDP multi-GPU training. Unpublished and preliminary.

EXPERIENCE

Computational Scientist

Feb 2018 – present

Center for Applied Scientific Computing, Lawrence Livermore National Laboratory

- Co-developed **PIAFS**, a high-order nonlinear hydrodynamics code with photochemistry for plasma-based gaseous optics, including the CUDA implementation (*Phys. Plasmas* 2025, *Phys. Rev. Applied* 2026).
- Developed **EUCLID**, a high-order multifluid solver for interpenetrating plasmas in inertial confinement fusion, under the programme I led as Principal Investigator.
- Contributed implicit particle-in-cell and Coulomb-collision algorithms to **PICNIC**, and high-order implicit-explicit time integration to **COGENT** for tokamak edge simulation.

Postdoctoral Research Staff Member

Oct 2015 – Feb 2018

Center for Applied Scientific Computing, Lawrence Livermore National Laboratory

- Implicit-explicit time integration for continuum kinetic simulation of collisional magnetized plasmas, making previously intractable regimes affordable.

Postdoctoral Appointee (and Fellow, Computation Institute, U. Chicago)

Feb 2013 – Oct 2015

Mathematics and Computer Science Division, Argonne National Laboratory

- Contributed implicit, explicit, and IMEX Runge-Kutta integrators and general linear methods with global error estimation to PETSc, used today across the computational science community.
- Removed the serial bottleneck in nonlinear compact finite-difference schemes with a distributed tridiagonal solver, demonstrating scaling on DOE leadership-class supercomputers (*SIAM J. Sci. Comput.*, 2015).

Graduate Research Assistant

Aug 2008 – Jan 2013

Alfred Gessow Rotorcraft Center, University of Maryland

- Derived and analysed the **CRWENO** schemes — compact reconstruction with weighted ENO limiting, raising spectral resolution at fixed formal order. My most-cited work; the schemes are used in several production codes.

TECHNICAL SKILLS

Languages	C++, C, Python, CUDA, MATLAB
HPC	MPI, OpenMP, CUDA, HIP, SYCL, AMReX, DOE leadership-class systems; GPU portability and performance
Numerics	PETSc (SNES/KSP/TS), SUNDIALS, Chombo, hypre; Newton-Krylov solvers, preconditioning, high-order discretization, implicit time integration
Machine learning	PyTorch, distributed data-parallel training, U-Net and encoder-decoder architectures, diffusion models, physics-informed objectives (<i>working proficiency; applied in current exploratory work</i>)
Tooling	Git, CMake, Spack, pytest, CI pipelines, Linux
AI-assisted dev	Claude Code, OpenAI Codex, Cursor — used daily across the projects above
Recent training	Generative AI (IPAM, UCLA, 2026) AI for Science (NERSC, 2025) Reinforcement Learning (2024)

SELECTED PUBLICATIONS (29 journal articles; full list at debog.github.io/publications)

- de Jong, E. K., Gunawardena, N., Katona, J. E., Beydoun, H., **Ghosh, D.**, Caldwell, P., *Data-Driven Reduced Order Modeling for Warm Rain Microphysics*, Journal of Geophysical Research: Machine Learning and Computation, 3 (3), 2026, e2025JH001103, [doi:10.1029/2025JH001103](https://doi.org/10.1029/2025JH001103).
- Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J. L., *Local reduced-order modeling for electrostatic plasmas by physics-informed solution manifold decomposition*, Computer Physics Communications, 322, 2026, 110039, [doi:10.1016/j.cpc.2026.110039](https://doi.org/10.1016/j.cpc.2026.110039).
- Oudin, A., **Ghosh, D.**, Riconda, C., Lancia, L., Kur, E., Ou, K., Perez-Ramirez, V. M., Lee, J., Edwards, M. R., Michel, P., *PIAFS: A 2D nonlinear hydrodynamics code to model gaseous optics*, Physics of Plasmas, 32, 2025, 072714, [doi:10.1063/5.0268318](https://doi.org/10.1063/5.0268318).
- Ghosh, D.**, Chapman, T. D., Berger, R. L., Dimits, A., Banks, J. W., *A Multispecies, Multifluid Model for Laser-Induced Counterstreaming Plasma Simulations*, Computers and Fluids, 186, 2019, 38-57, [doi:10.1016/j.compfluid.2019.04.012](https://doi.org/10.1016/j.compfluid.2019.04.012).
- Ghosh, D.**, Constantinescu, E. M., Brown, J., *Efficient Implementation of Nonlinear Compact Schemes on Massively Parallel Platforms*, SIAM Journal on Scientific Computing, 37 (3), 2015, C354–C383, [doi:10.1137/140989261](https://doi.org/10.1137/140989261).
- Ghosh, D.**, Baeder, J. D., *Compact Reconstruction Schemes with Weighted ENO Limiting for Hyperbolic Conservation Laws*, SIAM Journal on Scientific Computing, 34 (3), 2012, A1678–A1706, [doi:10.1137/110857659](https://doi.org/10.1137/110857659).

RESEARCH FUNDING

- **Principal Investigator** — Interpenetrating Plasma Simulations, LLNL Laboratory Directed Research and Development (LDRD), ~\$650k/yr, 2017–2020. Led a multi-institution team; produced a new multifluid capability and five journal articles.
- **Co-Investigator** — High-energy-density plasma kinetic simulation (LDRD, ~\$650k/yr, 2023–2026); phase-space methods in complex geometries (DOE ASCR, ~\$900k/yr, 2017–2024); implicit PIC/Monte-Carlo (LDRD, 2021–2022); sparse grids (LDRD, 2019–2020).

EDUCATION

Ph.D., Applied Mathematics and Scientific Computation — University of Maryland, College Park, 2013

B.Tech. and M.Tech. (dual degree), Aerospace Engineering — Indian Institute of Technology Bombay, 2006